

Cristhian Mallqui

Robot Learning | Perception | Autonomous Systems

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Profile

KTH MSc student and Research Engineer working on robot learning, visual scene representations and autonomous systems. Experience in mobile manipulation, perception, estimation and control; first-author publication in soft robotics. Seeking a robotics MSc thesis from January to July/August 2027 or a research internship, with interest in continuing in a research or engineering role.

Education

KTH Royal Institute of Technology

Aug 2025 – Present

MSc in Systems, Control and Robotics | Stockholm, Sweden

Autonomous Systems specialization; expected completion in 2027. Study includes robot learning, reinforcement learning, deep learning, computer vision and applied estimation.

Pontificia Universidad Católica del Perú

Mar 2018 – Dec 2023

BSc in Mechatronics Engineering | Lima, Peru

Top 10% of graduating class. Thesis: teleoperated inspection vehicle for weld seams and corrosion mapping on hard-to-reach ferromagnetic surfaces.

Lund University

Jan – Jun 2023

Exchange studies in Mechatronics | Lund, Sweden

Projects in real-time control systems and mechatronic industrial product design.

Research Experience

Research Engineer | KTH Robotics, Perception and Learning

Oct 2025 – Present

ALARS project | Supervisors: John Folkesson and Özer Özkahraman

- Develop perception for autonomous aerial recovery of underwater robots; calibrate a 360-degree camera and train YOLOv11 detectors using real and synthetic imagery.
- Work on a DJI Matrice 350 landing pipeline that connects oriented object detections, EKF relative-state estimation, a landing state machine and LQR control in ROS 2.
- Develop and debug trajectory tracking and landing logic in simulation, including quintic references, input shaping and handling of temporary perception loss.

Summer Research Assistant | University of Delaware

Jul – Nov 2024

Soft robotics | Supervisor: Nathan Lazarus

- Led two projects on liquid-metal vibrotactile actuators and magnetic adhesion systems, combining prototype design, fabrication, performance characterization and safety analysis.
- Validated wearable prototypes through user testing; research resulted in a first-author journal publication and an invention disclosure in progress.

Current Robotics Projects

Mobile Manipulation and Imitation Learning | RBY1 / MoBiPi

Ongoing

KTH engineering project | Rainbow Robotics RBY1

- Develop a simulation and demonstration-processing pipeline using MuJoCo, robosuite, RoboCasa, robomimic and MimicGen for manipulation tasks, including door closing.
- Debug controller metadata and action representations; validate an action adapter on 100 saved simulator states, including contact and late-stage pushing configurations.
- Train and evaluate Diffusion Policy using RGB and end-effector observations; investigate demonstration replay, generated-trajectory failures and policy rollout behavior.

RGB Room Transitions and Topological Scene Representation

Ongoing

- Develop room-transition detection using RADIO embeddings and temporal convolutional networks; evaluate on indoor Spot/Odin rosbag sequences and investigate missed transitions and repeated peaks.
- Build Habitat-Sim trajectory collection for HM3D and develop room-level visual memory and candidate-room matching toward a topological representation without a globally consistent metric map.

Selected Engineering Projects

Long-Term Dynamic Mapping | OneMap Extension

Jan 2026 – Present

KTH | Supervisors: Olov Andersson, Timon Homberger and Finn Busch

- Extend foundation-model semantic mapping toward multi-floor and changing environments; integrate Odin 1 point-cloud fusion and evaluate dynamic-object handling.

Vehicle State Estimation | KTH Formula Student

Oct 2025 – Jan 2026

- Developed a cascaded UKF–EKF architecture for vehicle velocity and attitude estimation under wheel slip, using adaptive filtering and innovation gating; tested on real vehicle data at 100 Hz.

Illaryi Exploration Rover | European Rover Challenge

Feb – Aug 2025

- Implemented multi-mode steering and a ROS 2 autonomy stack on Jetson Orin NX, integrating RealSense D455 and Livox MID-360 sensing, visual SLAM, sensor fusion and nvblox costmaps.

Publications

Liquid Touch: Soft and Wearable Vibrotactile Actuators Based on Room Temperature Liquid Metals.

Advanced Engineering Materials, 2026. First-author journal paper.

Gone Fishing: Post-Mission Autonomous Aerial Recovery of Unmanned Underwater Vehicles. *IEEE OES AUV Symposium*, 2026. Accepted extended abstract; contribution to underwater perception for ALARS.

Robotics Summer Schools

ETH RobotX Robotics Summer School

2026

- Tuned exploration and state-estimation parameters and worked on object-detection pipelines with persistent detections in the provided ROS 2 autonomy stack.
- Tested the pipeline in MuJoCo and transferred it to a real Unitree A2 quadruped.

RimA Summer School | FZI and KIT

3–7 Aug 2026

- Developed shoe detection and classification, including recognition of unknown shoes, using vision foundation models and YOLO for the Unitree Go2.
- Implemented behavior trees on Boston Dynamics Spot for navigation and manipulation tasks in indoor environments.

Additional Experience and Leadership

Research Intern | PUCP Core Facilities – FABCORE

Aug 2023 – Feb 2024

Designed and implemented a 1.5 m clay delta 3D printer across mechanics, electronics, software and control; trained new interns and supported 3D-printing services.

University Teaching Assistant | PUCP

Mar 2024 – Aug 2025

Taught robotics and AI topics from kinematics to motion planning, mechatronic project design and physics; supported courses across engineering and computer science.

Leadership and Mentoring | PUCP Robotics Group

Nov 2018 – Aug 2025

Served as President, Vice President and Workshop Coordinator; led technical programs and mentored robotics competition teams.

Technical Skills

Programming and development: Python, C/C++, MATLAB, Linux, Git, Docker, CMake.

Robot learning and vision: PyTorch, OpenCV, imitation learning, Diffusion Policy, YOLO, RADIO, SAM; reinforcement learning and VLA concepts.

Robotics: ROS 2, micro-ROS, MoveIt 2, ros2_control, Nav2, SLAM, EKF/UKF sensor fusion, LQR.

Simulation and data: MuJoCo, robosuite, robomimic, MimicGen, RoboCasa, Habitat-Sim, Isaac Sim, Unity, HDF5, Rerun.

Hardware and design: Jetson, Raspberry Pi, STM32, RealSense, LiDAR; SolidWorks, Fusion 360.

Awards and Languages

Team awards: Latin American Space Challenge: CanSat first place (2021, 2023) and Satellite Challenge overall winner (2023). European Rover Challenge Remote Edition: first in Latin America and 11th overall (2023).

Languages: Spanish (native); English.